

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

ĐµĐ¶Đµ ÑĤĐ°Đ°, Đ´Đ°Đ²Đ°Đ¹ÑĤĐµ ÑĢĐµÑĪĐ,Đ¼ Đ·Đ°Đ´Đ°ÑĭÑĥ Đ¿Đ³¼ÑĪĐ°Đ³Đ³¼Đ²Đ³¼.

ĐĤĐ±Đ³¼Đ·Đ¹²Đ°ÑĭĐ,Đ¼,ÑģĐ°Đ³¼Đ»ÑĹĐ°Đ³¼Đ²Đ°Đ°ÑĥÑĥĐ¹¼-Đ¿ÑĢĐ,Đ½Đ°Đ´Đ»ĐµĐ¶Đ½Đ³¼ÑģÑĤĐµĐ¹ÑĥĐĲĐµĐ»Đ°Đ½Đ,Đ²Đ½Đ°ÑĭĐ°Đ»ĐµâĢĶÑĹÑĤĐ³¼ÑĭĐ,ÑģĐ»Đ³¼**x**.

1. **ĐĹĐµĐ»ĐµĐ½ÑĭĐ¹Đ´Đ³¼Đ¹¼** : Đ³¼Đ½Đ°Đ¿ÑĢĐ³¼Đ´Đ°Đ»Đ°ÑĤÑĢĐµÑĤÑĹĐ²Đ°Đ°ÑĥÑĥĐ¹¼Đ³¼Đ².

âĲĐ¿ÑĢĐ³¼Đ´Đ°Đ»Đ° : x / 3
âĲĐ³¼ÑģÑĤĐ°Đ»Đ³¼ÑģÑĹ : x - x / 3 = (2 x) / 3

2. **ĐĲÑĢĐ°ÑģĐ½ÑĭĐ¹Đ´Đ³¼Đ¹¼** : Đ¿ÑĢĐ³¼Đ´Đ°Đ»Đ° 2 Đ±Đ³¼Đ»ÑĹÑĪĐµ,ÑĭĐµĐ¹¼Đ²Đ·ĐµĐ»ĐµĐ½Đ³¼Đ¹¼Đ´Đ³¼Đ¹¼Đµ

ĐĴĐ·ĐµĐ»ĐµĐ½Đ³¼Đ¹¼Đ´Đ³¼Đ¹¼ĐµĐ³¼Đ½Đ°Đ¿ÑĢĐ³¼Đ´Đ°Đ»Đ° x / 3 , Đ·Đ½Đ°ÑĭĐ,ÑĤ,Đ²Đ°ÑĢĐ°ÑģĐ½Đ³¼Đ¹¼Đ´Đ³¼Đ¹¼ĐµĐ¿ÑĢĐ³¼Đ´Đ°Đ»Đ° x / 3 + 2

âĲĐ³¼ÑģÑĤĐ°Đ»Đ³¼ÑģÑĹ :
(2 x) / 3 - (x / 3 + 2) = (2 x) / 3 - x / 3 - 2 = (x / 3) - 2

3. **ĐĲÑĢĐ°Đ½Đ¶ĐµĐ²ÑĭĐ¹Đ´Đ³¼Đ¹¼** : Đ¿ÑĢĐ³¼Đ´Đ°Đ»Đ°**Đ¿Đ³¼Đ»Đ³¼Đ²Đ,Đ½ÑĥĐ³¼ÑģÑĤĐ°Đ²ÑĪĐ,ÑĥÑģÑĲ**Đ²Đ°Đ°ÑĥÑĥĐ¹¼Đ³¼Đ².

âĲĐ¿ÑĢĐ³¼Đ´Đ°Đ»Đ° : (1 / 2) ÃĹ [(x / 3) - 2]
âĲĐ³¼ÑģÑĤĐ°Đ»Đ³¼ÑģÑĹ : (1 / 2) ÃĹ [(x / 3) - 2]

ĐŁĐ³¼ÑĥÑģĐ»Đ³¼Đ²Đ,ÑĲ,Đ³¼ÑģÑĤĐ°Đ»Đ³¼ÑģÑĹ**5**Đ²Đ°Đ°ÑĥÑĥĐ¹¼Đ³¼Đ².

ĐĸĐ°Đ°ÑĭÑĤĐ³¼ :
\$\$
\frac { 1 }{ 2 } \left(\frac { x }{ 3 } - 2 \right) = 5
\$\$

ĐĲĐµÑĪĐ,Đ¼ÑĹÑĤĐ³¼ÑĥÑĢĐ°Đ²Đ½ĐµĐ½Đ,Đµ :

Đ£Đ¹¼Đ½Đ³¼Đ¶Đ,Đ¼Đ³¼Đ±ĐµÑģÑĤĐ³¼ÑĢĐ³¼Đ½ÑĭĐ½Đ° 2 :
\$\$
\frac { x }{ 3 } - 2 = 10
\$\$

ĐŁÑĢĐ,Đ±Đ°Đ²Đ,Đ¼2 :
\$\$
\frac { x }{ 3 } = 12
\$\$

Đ£Đ¹¼Đ½Đ³¼Đ¶Đ,Đ¼Đ½Đ° 3 :
\$\$
x = 36
\$\$

ĐĲÑĤĐ²ĐµÑĤ : ĐĲĐµĐ»Đ°Đ½Đ,Đ½Đ°ÑĭĐ°Đ»Đ°Ñģ36Đ²Đ°Đ°ÑĥÑĥĐ¹¼-Đ¿ÑĢĐ,Đ½Đ°Đ´Đ»ĐµĐ¶Đ½Đ³¼ÑģÑĤĐµĐ¹.

< answer > 36 </answer>